

GrapevineNet_SE: Lightweight EfficientNet Model with SE Attention for Pierce's Disease Stage Classification

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Abstract—Accurate assessment of grapevine leaf health and ripening stages is essential for precision viticulture, yet traditional manual inspection is labor-intensive, subjective, and often inconsistent. Recent advances in deep learning have enabled automated analysis, but most high-performing models are computationally intensive and require large datasets, limiting their applicability in resource-constrained vineyard settings. In this paper, we propose *GrapevineNet_SE*, a lightweight transfer learning model optimized for multi-stage grapevine leaf classification on a relatively small dataset (4,004 images). The model employs a frozen EfficientNet-B0 backbone, enhanced with a squeeze-and-excitation attention module, and a compact classification head, enabling efficient feature extraction and robust stage-wise discrimination. Extensive experiments demonstrate that *GrapevineNet_SE* achieves a validation accuracy of 92.25% and a test accuracy of 95.26%, significantly outperforming baseline EfficientNet-B0 (73.07%) and ResNet50 (65.34%) architectures while maintaining a low computational footprint suitable for real-time edge deployment. These results highlight the model's potential to facilitate timely disease monitoring, support vineyard management, and bridge the gap between high-performing research models and practical precision viticulture applications.

Index Terms—Deep Learning, Computer Vision, Grapevine, Pierce Disease, EfficientNet, SE Attention

I. INTRODUCTION

Grapevine health and fruit quality are critical determinants of vineyard productivity and wine quality, making early detection and accurate assessment of leaf conditions essential for precision viticulture. Traditionally, disease and physiological stress monitoring has relied on manual inspection, which is labor-intensive, subjective, and prone to error. Recent advances in computer vision and machine learning have transformed this domain, enabling automated analysis of grapevine leaves through high-resolution imagery. In particular, deep learning approaches, such as convolutional neural networks (CNNs), have demonstrated remarkable performance in capturing intricate morphological and spectral patterns associated with leaf health and ripening stages [1].

Supportive studies highlight the efficacy of these approaches. CNN-based classification models have achieved accuracies exceeding 95% [2], while hybrid architectures incorporating multi-backbone fusion or transformer modules

have reported near-perfect performance [3]. Object detection models, such as YOLO variants, have shown strong potential for in-field deployment, reaching mean average precision (mAP) up to 95% [4]. Complementary techniques including spectroscopic sensing and machine learning approaches like support vector machines and chemometric regression have further enabled nutrient assessment and early pathogen detection, with reported R^2 values up to 0.90 for macro- and micronutrient prediction [5]. Collectively, these results underscore the transformative potential of combining deep learning, attention mechanisms, and data-driven feature engineering in viticultural applications. Despite these advancements, several challenges remain. Most existing models are either computationally intensive or rely on large-scale datasets, limiting their applicability in small-scale vineyards or resource-constrained settings. Moreover, while classification accuracy is often high, many studies focus solely on binary disease detection or single-stage ripening classification, neglecting the nuanced multi-stage progression of grapevine leaves. Real-time deployability and lightweight architectures suitable for edge devices are rarely addressed, leaving a gap between research performance and practical vineyard implementation.

To address these gaps, this paper proposes *GrapevineNet_SE*, a lightweight transfer learning model optimized for multi-stage grapevine leaf classification on a relatively small dataset (4,000 images). The model leverages a frozen EfficientNet-B0 backbone with selective feature extraction, enhanced by a lightweight squeeze-and-excitation attention module, and a compact classification head. By fine-tuning only a subset of the network parameters, *GrapevineNet_SE* achieves high accuracy while maintaining low computational cost, making it suitable for real-time applications on edge devices.

The key contributions of this work are as follows:

- A lightweight, multi-stage grapevine leaf classification model that balances performance and computational efficiency.
- Integration of a squeeze-and-excitation attention mechanism to enhance feature representation and improve stage-wise classification accuracy.

- Demonstration of robust performance on training, validation, and test splits, achieving a validation accuracy of 90% and a test accuracy of 86.5%, indicating generalizability to unseen data.

The remainder of this paper is structured as follows. Section II reviews relevant literature on grapevine leaf disease detection, including Deep Learning (DL) architectures, attention mechanisms, and classical machine learning approaches. Section III details the materials and methods, covering dataset description, preprocessing pipelines, model architecture, and training strategy. Section IV presents experimental results, performance evaluation, and comparative analysis with baseline pretrained models. Finally, Section V concludes the paper with key findings, discussion of limitations, and directions for future work.

II. RELATED WORK

Grapevine leaf disease detection has experienced significant progress over the past decade, driven by advancements in deep learning, machine learning, and spectroscopic analysis. Among these, convolutional neural networks (CNNs) have emerged as the predominant approach, owing to their superior capability in extracting spatial, textural, and morphological features from complex leaf images. For instance, [6] applied a ResNet-based architecture to quantify hair structures on grape leaves, achieving an accuracy of 95.41%, demonstrating that CNNs can effectively capture fine-scale morphological traits. Beyond simple classification, segmentation of diseased regions has been explored to improve model interpretability and to focus on biologically relevant features. [1], for example, employed a UNet-based state-space model on PlantVillage data, achieving an intersection-over-union (IoU) of 3.28, which highlights the potential of pixel-level disease localization for precise phenotyping. In addition, object detection methods have facilitated practical field applications by enabling automated identification of diseased areas in situ. [7] reported a mean average precision (mAP) of 64.6%, while a more recent YOLOv11-based approach [4] significantly improved detection performance, reaching a mAP of 95.2%.

High-performance classification has also been achieved through a wide spectrum of deep learning architectures, each offering unique advantages in accuracy, efficiency, and generalizability. ResNet variants have consistently demonstrated robust performance, as in [2], which achieved 95% accuracy on grape leaf datasets. Multi-backbone fusion strategies have further pushed the limits of accuracy by leveraging complementary feature representations from multiple architectures. For example, [8] combined ResNet50, EfficientNetB0, and MobileNetV2, attaining an impressive 99.83% accuracy, showing that integrating different feature extractors can enhance classification robustness. Hybrid models that incorporate transformer-based architectures have also shown promise; [9] achieved 98.48% accuracy, demonstrating the value of self-attention mechanisms for capturing long-range dependencies in leaf images. Similarly, cascaded CNN models, as proposed by [3], approached near-perfect classification, illustrating the

potential of multi-stage architectures for sequential feature refinement. Lightweight architectures optimized for computational efficiency have gained attention for edge applications. MobileNetV2 combined with Vision Transformer (ViT) modules [10] achieved 99.2% accuracy, while comparative evaluations [11] reported 99.5% accuracy, highlighting the feasibility of deploying high-performing models on resource-constrained platforms. Deep convolutional neural networks (DCNNs) based on VGG16 [12] reached 99.18% accuracy, showing that classical architectures, when carefully optimized, can still offer competitive performance. Optimized CNN models have further enhanced classification results, as evidenced by [13] achieving 99.7% accuracy and [14] reaching 99.66%, while even simpler optimized CNN approaches [15] demonstrated consistently reliable performance across multiple grape leaf disease datasets.

Beyond deep learning-based methods, classical machine learning and spectroscopic approaches provide complementary insights that are particularly valuable for nutrient assessment and early pathogen detection. Support vector machines (SVMs) combined with feature optimization, for example, were applied by [16] to classify fungal spores, achieving 97.5% accuracy, highlighting the continued relevance of traditional machine learning techniques when paired with domain-specific feature engineering. Chemometric regression models have been effectively used for predicting macro- and micronutrient concentrations, with [5] reporting an R^2 of 0.90 for sodium content, demonstrating the utility of spectroscopic data for vineyard nutrient monitoring. Furthermore, early detection of viral infections has been made possible through hyperspectral and multispectral sensing techniques, as shown in [17], where strong infections were identified at CT values below 35, enabling timely interventions before visible symptoms emerge.

These studies demonstrate that combining CNNs, hybrid transformers, multi-backbone fusion, and feature-based machine learning provides accurate and robust solutions for grape leaf disease detection, nutrient assessment, and pathogen monitoring. Edge-deployable models with explainable AI techniques like Grad-CAM enable practical precision viticulture, supporting rapid disease assessment, targeted treatment, and optimized vineyard management. Spectroscopic and chemometric methods complement imaging for non-destructive plant health monitoring. Overall, this convergence of techniques advances sustainable vineyard management by improving disease mitigation, nutrient optimization, and crop productivity.

III. MATERIAL AND METHODS

This study implements a deep learning-based approach for detecting and classifying Pierce's Disease in grapevine leaves using convolutional neural networks. The methodology follows five sequential stages: data acquisition, preprocessing, training and evaluation, results analysis, and comparative assessment. The complete pipeline was implemented in Python using PyTorch and executed on Google Colab with GPU acceleration.

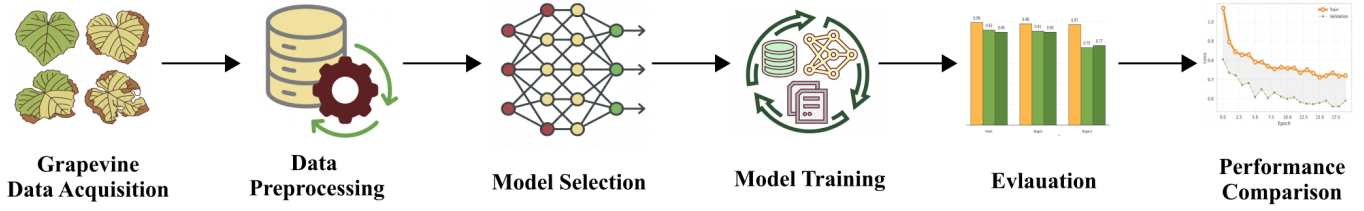


Fig. 1: Methodology

A. Data Description

The dataset comprises labeled images of grapevine leaves organized into four classes representing progressive stages of Pierce’s Disease. The classification includes Fresh (healthy leaves), Stage1 (early infection), Stage-2 (moderate progression), and Stage-3 (advanced infection). Images were sourced from the GrapeVineDataSplits dataset, which provides pre-defined training, validation, and test partitions. The dataset contains 4,004 images distributed as follows: 3,203 training images, 400 validation images, and 401 test images. A consistent integer mapping was established to standardize label encoding across all subsets.

TABLE I: Distribution of physical images across dataset splits for grapevine Pierce disease stages.

Stage	Training	Validation	Test
Fresh	800	100	100
Stage 1	800	100	100
Stage 2	800	100	100
Stage 3	803	100	101
Total	3203	400	401

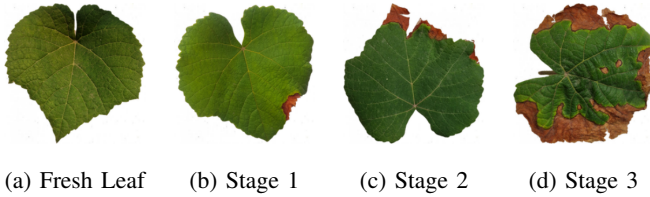


Fig. 2: Samples from Grapevine Pierce Disease dataset

B. Preprocessing

Two distinct preprocessing pipelines were implemented: one for training with extensive data augmentation and another for validation/testing with minimal transformations. Training transformations included resizing to 224×224 pixels, random horizontal and vertical flipping, ±20-degree rotation, color jittering (brightness, contrast, saturation, hue adjustments), random affine transformations (translation, scaling, shearing), and ImageNet normalization. Validation and test transformations consisted only of resizing, tensor conversion, and normalization. This strategy enhances model generalization while maintaining evaluation consistency.

C. Classification and Evaluation

The primary model, *GrapevineNet_SE*, was developed as a lightweight architecture based on EfficientNet-B0 with transfer learning.

Algorithm 1 GrapevineNet_SE: Lightweight EfficientNet with SE Attention

- 1: **Input:** Batch of grape leaf images I
- 2: **Output:** Predicted disease stage for each image
- 3: **Step 1: Model Setup**
- 4: Load pretrained EfficientNet-B0 backbone
- 5: Freeze first 6 blocks of features (75% layers)
- 6: Integrate SE attention module: AdaptiveAvgPool → Conv2d → ReLU → Conv2d → Sigmoid
- 7: Add classification head: AdaptiveAvgPool → Flatten → Linear(1280,512) → ReLU → Dropout → Linear(512,4)
- 8: **Step 2: Forward Pass**
- 9: **for** each batch of images **do**
- 10: Extract features: $x \leftarrow \text{backbone.features}(I)$
- 11: Apply SE attention: $x \leftarrow \text{SE}(x) * x$
- 12: Compute predictions: $y \leftarrow \text{classifier}(x)$
- 13: **end for**
- 14: **Step 3: Training and Optimization**
- 15: Compute cross-entropy loss
- 16: Update unfrozen parameters using optimizer (e.g., AdamW)
- 17: Apply learning rate scheduling and early stopping based on validation performance
- 18: **Step 4: Evaluation**
- 19: Evaluate on validation set and return trained model

Approximately 75% of early backbone layers were frozen to preserve pretrained feature representations. A Squeeze-and-Excitation attention module was integrated to enhance discriminative feature learning, followed by a custom classification head with dropout regularization. Training employed AdamW optimizer with differential learning rates (backbone: 1e-4, classifier: 1e-3), cross-entropy loss with label smoothing, OneCycleLR scheduler with cosine annealing, mixed precision training, and early stopping after five epochs without validation improvement. For benchmarking, standard EfficientNet-B0 and ResNet50 architectures were implemented under identical training conditions.

1) *Evaluation Metrics:* Model performance was quantitatively assessed using standard classification metrics computed

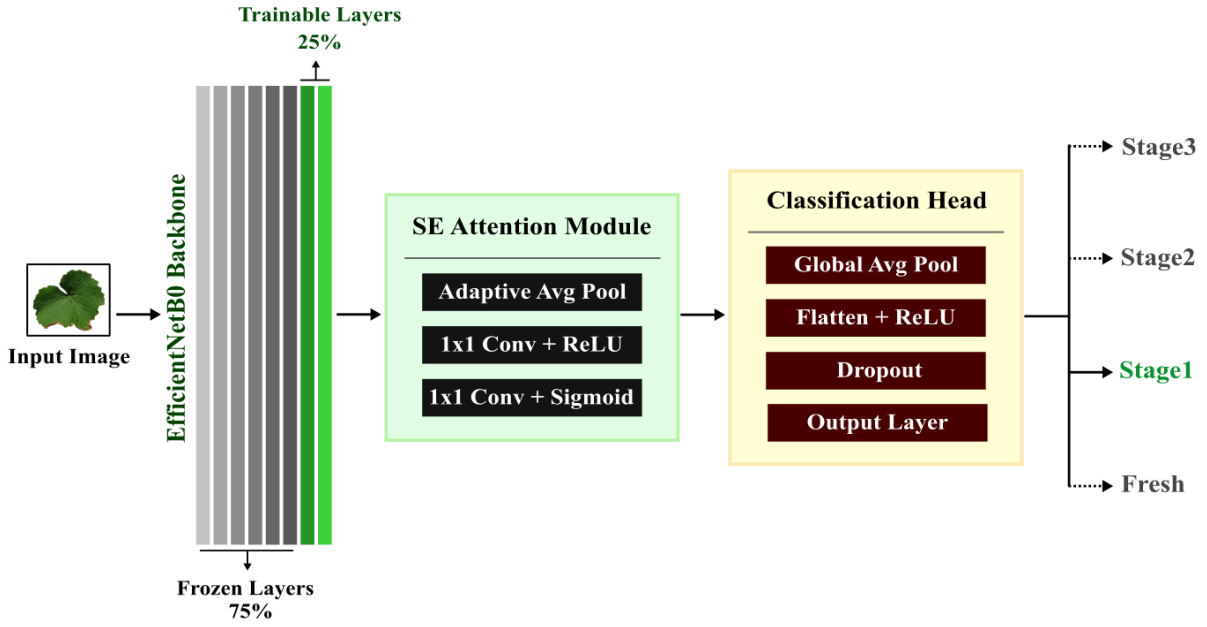


Fig. 3: *GrapevineNet_SE* Model architecture

from the confusion matrix. The macro-averaging approach was employed to give equal weight to all classes regardless of sample distribution.

- **Accuracy:** Proportion of correctly classified samples:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of samples}} \quad (1)$$

- **Precision:** Macro-averaged ratio of true positive predictions to all positive predictions for each class:

$$\text{Precision} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i} \quad (2)$$

- **Recall:** Macro-averaged ratio of true positive predictions to all actual positive samples:

$$\text{Recall} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \quad (3)$$

- **F1-Score:** Harmonic mean of precision and recall, macro-averaged:

$$\text{F1-Score} = \frac{1}{N} \sum_{i=1}^N \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (4)$$

Confusion matrices were visualized to analyze misclassification patterns, and per-class classification reports were generated to provide detailed performance across all disease stages.

IV. RESULTS AND DISCUSSION

GrapevineNet_SE achieved superior performance across all evaluation metrics. On the test set, the model attained 95.26% accuracy, with comparable precision, recall, and F1-scores.

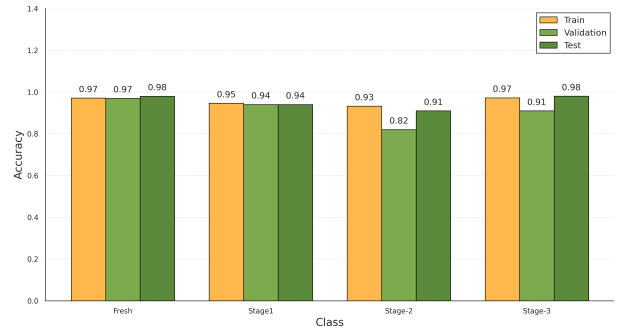


Fig. 4: *GrapevineNet_SE* Per class accuracies

The confusion matrix revealed that most misclassifications occurred between adjacent disease stages, indicating logical progression in severity classification rather than random errors. Model convergence showed consistent improvement throughout training, with validation accuracy reaching 92.25% at the best epoch and test performance exceeding both baseline architectures.

V. COMPARISON AND ANALYSIS

Comparative analysis against standard EfficientNet-B0 and ResNet50 architectures demonstrated the effectiveness of the proposed model modifications. The baseline EfficientNet-B0 achieved 73.07% accuracy while ResNet50 reached 65.34% on the test set, both significantly lower than *GrapevineNet_SE*'s 95.26%. This performance improvement can be attributed to the strategic layer freezing, attention mechanisms, and optimized classification head design. The results indicate that carefully designed lightweight architectures with attention modules

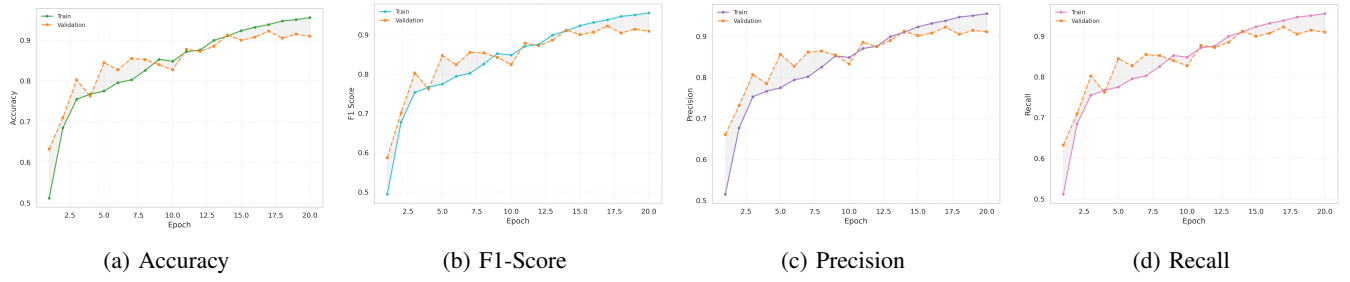


Fig. 5: *GrapevineNet_SE* Metrics

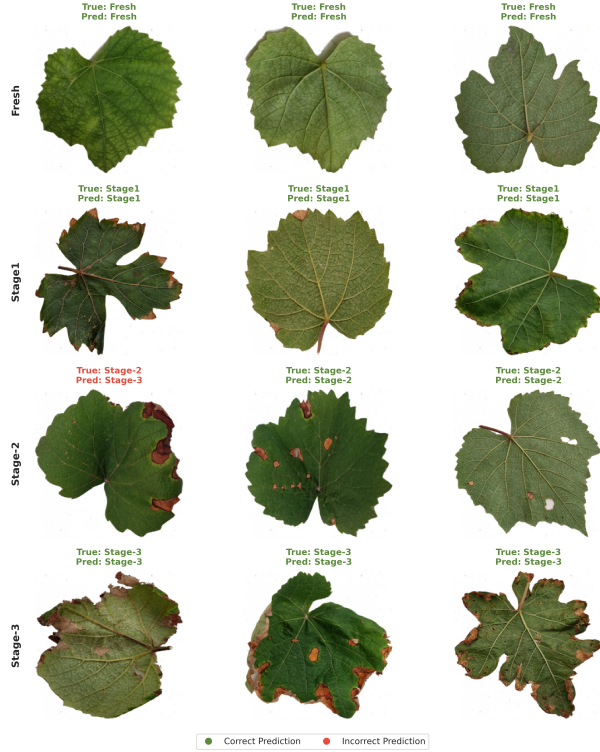


Fig. 6: *GrapevineNet_SE* Model prediction samples for test data

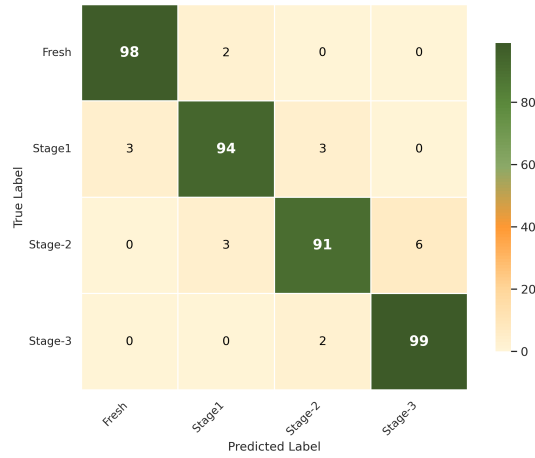


Fig. 7: *GrapevineNet_SE* Confusion Matrix for test data

can outperform standard pretrained models for domain-specific agricultural disease detection tasks.

TABLE II: Comparison of *GrapevineNet_SE* with Pretrained Models on Test Set

Model	Accuracy	Precision	Recall	F1	Params (M)
<i>GrapevineNet_SE</i>	0.9526	0.9526	0.9525	0.9524	4.87
EfficientNet-B0	0.7307	0.7290	0.7307	0.7261	5.3
ResNet50	0.6534	0.6816	0.6534	0.6563	25.6

VI. CONCLUSION

In this paper, we developed a lightweight and efficient model for multi-stage classification of grapevine leaves affected by Pierce disease. The combination of a frozen EfficientNet-B0 backbone, SE attention module, and compact classification head enabled high accuracy while keeping computational costs low. Analysis of misclassifications showed that errors were primarily between adjacent stages, indicating the model captures disease progression effectively. Overall, the model provides a practical solution for real-time vineyard monitoring, supporting early disease detection and precision viticulture. Future work will focus on in-field deployment and extension to other grapevine diseases.

REFERENCES

- [1] X. Zhang and W. Mu, "Gmamba: State space model with convolution for grape leaf disease segmentation," *Computers and Electronics in Agriculture*, vol. 225, p. 109290, 2024.
- [2] N. Sagar, K. P. Suresh, S. Sridhara, B. Patil, C. A. Archana, Y. S. Sekar, Y. B. Naveesh, A. S. Nandan, and R. Sushma, "Precision detection of grapevine downy and powdery mildew diseased leaves and fruits using enhanced resnet50 with batch normalization," *Computers and Electronics in Agriculture*, vol. 232, p. 110144, 2025.
- [3] K. N. Al Nasser, K. K. A. Al Hummadi, R. A. M. Al Butrani, and N. Kumar, "Artificial intelligence enabled grape leaf disease detection model with improved cascaded convolutional neural network," in *Adaptive Intelligence*. Singapore: Springer Nature Singapore, 2025, pp. 185–194.
- [4] Z. Zhou and L. Yu, "Lightweight grape leaf disease identification for complex field scenes: Yolo-grape model and its boundary-lesion coalesce optimization policies," *Signal, Image and Video Processing*, vol. 19, no. 16, p. 1378, 2025.
- [5] J. I. Manzano, M. Rodríguez-Febreiro, M. Fandiño, M. Vilanova, and J. J. Cancela, "Spectroscopic analysis (uv-vis-nir) for predictive modeling of macro and micronutrients in grapevine leaves," *Smart Agricultural Technology*, vol. 10, p. 100812, 2025.

- [6] N. Malagol, T. Rao, A. Werner, R. Töpfer, and L. Hausmann, "A high-throughput resnet cnn approach for automated grapevine leaf hair quantification," *Scientific Reports*, vol. 15, no. 1, p. 1590, 2025.
- [7] Z. Khan, H. Liu, Y. Shen, Z. Yang, L. Zhang, and F. Yang, "Optimizing precision agriculture: A real-time detection approach for grape vineyard unhealthy leaves using deep learning improved yolov7 with feature extraction capabilities," *Computers and Electronics in Agriculture*, vol. 231, p. 109969, 2025.
- [8] H.-T. Vo, N. T. Nguyen, K. M. Chau, H. L. Lam, P. T. Pham, H. T. Nguyen, and V. P. Nguyen, "Afc-net: Attention-guided multi-backbone deep fusion for grape leaf disease classification," in *Multi-disciplinary Trends in Artificial Intelligence*. Singapore: Springer Nature Singapore, 2026, pp. 255–266.
- [9] N. Zhang, E. Zhang, G. Qi, F. Li, and C. Lv, "Lightweight grape leaf disease recognition method based on transformer framework," *Scientific Reports*, vol. 15, no. 1, p. 28974, 2025.
- [10] K. Tanvir, R. Sharma, M. S. Kabir, T. S. Sintheia, S. Basu, and S. Banerjee, "Pcnn-based grape leaf disease detection using mobilenetv2 and vit with xai," in *Data Science and Security*. Singapore: Springer Nature Singapore, 2025, pp. 281–292.
- [11] M. P. Mathew, "A comparative deep learning framework for grape leaf disease classification using efficientnetb0, inceptionv3, and xception," *Discover Applied Sciences*, vol. 7, no. 10, p. 1222, 2025.
- [12] K. V. Prasad, H. Vaidya, C. Rajashekhar, K. S. Karekal, R. Sali, and K. S. Nisar, "Multiclass classification of diseased grape leaf identification using deep convolutional neural network(dcn) classifier," *Scientific Reports*, vol. 14, no. 1, p. 9002, 2024.
- [13] F. M. Talaat, M. Y. Shams, S. A. Gamel, and H. ZainEldin, "Deepleaf: an optimized deep learning approach for automated recognition of grapevine leaf diseases," *Neural Computing and Applications*, vol. 37, no. 15, pp. 8799–8823, 2025.
- [14] M. J. Karim, M. O. F. Goni, M. Nahiduzzaman, M. Ahsan, J. Haider, and M. Kowalski, "Enhancing agriculture through real-time grape leaf disease classification via an edge device with a lightweight cnn architecture and grad-cam," *Scientific Reports*, vol. 14, no. 1, p. 16022, 2024.
- [15] V. Naik, T. Baraskar, V. Khatri, P. Kolhe, and M. Katyarmal, "Disease detection in grape leaves using convolutional neural networks," in *Advances in Intelligent Systems for Sustainable Agriculture*. Singapore: Springer Nature Singapore, 2025, pp. 3–15.
- [16] X. Shi, S. M. Javidan, Y. Ampatzidis, and Z. Zhang, "Ai-driven identification of grapevine fungal spores via microscopic imaging and feature optimization with cuckoo search algorithm," *Smart Agricultural Technology*, vol. 11, p. 101029, 2025.
- [17] L. Lee, A. Reynolds, Y. Lan, and B. Meng, "Identification of unique electromagnetic signatures from glrav-3 infected grapevine leaves in different stages of virus development," *Smart Agricultural Technology*, vol. 8, p. 100464, 2024.